1. ip address show or ip a

```
student@fedora:~$ ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:2d:91:bf brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 86231sec preferred_lft 86231sec
    inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac/64 scope global dynamic noprefixroute
       valid_lft 86234sec preferred_lft 14234sec
    inet6 fe80::55f0:9771:cb55:7bf3/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

2. ip link show

```
student@fedora:~$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:2d:91:bf brd ff:ff:ff:ff:ff:ff
```

3. ip route show

```
student@fedora:~$ ip route show
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
```

4. ip neigh show

```
student@fedora:~$ ip neigh show
10.0.2.2 dev enp0s3 lladdr 52:55:0a:00:02:02 STALE
fe80::2 dev enp0s3 lladdr 52:56:00:00:00:02 router STALE
```

5. ping<hostname/IP>

```
student@fedora:~$ ping fedora
PING fedora(fedora (fe80::55f0:9771:cb55:7bf3%enp0s3)) 56 data bytes
64 bytes from fedora (fe80::55f0:9771:cb55:7bf3%enp0s3): icmp_seq=1 ttl=64 time=0.035 ms
64 bytes from fedora (fe80::55f0:9771:cb55:7bf3%enp0s3): icmp_seq=2 ttl=64 time=0.076 ms
64 bytes from fedora (fe80::55f0:9771:cb55:7bf3%enp0s3): icmp_seq=3 ttl=64 time=0.064 ms
64 bytes from fedora (fe80::55f0:9771:cb55:7bf3%enp0s3): icmp_seq=4 ttl=64 time=0.071 ms
64 bytes from fedora (fe80::55f0:9771:cb55:7bf3%enp0s3): icmp_seq=5 ttl=64 time=0.086 ms
64 bytes from fedora (fe80::55f0:9771:cb55:7bf3%enp0s3): icmp_seq=6 ttl=64 time=0.089 ms
64 bytes from fedora (fe80::55f0:9771:cb55:7bf3%enp0s3): icmp_seq=7 ttl=64 time=0.267 ms
64 bytes from fedora (fe80::55f0:9771:cb55:7bf3%enp0s3): icmp_seq=8 ttl=64 time=0.165 ms
64 bytes from fedora (fe80::55f0:9771:cb55:7bf3%enp0s3): icmp_seq=9 ttl=64 time=0.070 ms
64 bytes from fedora (fe80::55f0:9771:cb55:7bf3%enp0s3): icmp_seq=10 ttl=64 time=0.080 ms
64 bytes from fedora (fe80::55f0:9771:cb55:7bf3%enp0s3): icmp_seq=11 ttl=64 time=0.069 ms
```

6. hostname

```
student@fedora:~$ hostname
fedora
```

7. ss

```
student@fedora:~$ ss
Netid           State           Recv-Q          Send-Q                                  Local Address:Port                  Peer Address:Port          Process
u_str           ESTAB           0               0                                               * 23417                            * 23418
u_str           ESTAB           0               0                     /run/dbus/system_bus_socket 22134                            * 22133
u_str           ESTAB           0               0                     /run/dbus/system_bus_socket 17833                            * 18636
u_str           ESTAB           0               0                     /run/dbus/system_bus_socket 23629                            * 23454
u_str           ESTAB           0               0                                               * 20469                            * 20470
u_str           ESTAB           0               0                                               * 22368                            * 22369
u_str           ESTAB           0               0                                               * 21889                            * 22539
u_str           ESTAB           0               0                     /run/systemd/journal/stdout 19654                            * 19653
```

8. netstat

```
student@fedora:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 fedora:bootpc           _gateway:bootps         ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     23417
unix  3      [ ]         STREAM     CONNECTED     22134    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     17833    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     23629    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     20469
unix  3      [ ]         STREAM     CONNECTED     22368
```

9. ifconfig

```
student@fedora:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::55f0:9771:cb55:7bf3  prefixlen 64  scopeid 0x20<link>
        inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac  prefixlen 64  scopeid 0x0<global>
        ether 08:00:27:2d:91:bf  txqueuelen 1000  (Ethernet)
        RX packets 64  bytes 23284 (22.7 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 115  bytes 13057 (12.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

10. traceroute<hostname/IP>

```
student@fedora:~$ traceroute fedora
traceroute to fedora (10.0.2.15), 30 hops max, 60 byte packets
 1  fedora (10.0.2.15)  0.057 ms  0.008 ms  0.006 ms
```

11. nslookup<domain>

```
student@fedora:~$ traceroute fedora
traceroute to fedora (10.0.2.15), 30 hops max, 60 byte packets
 1  fedora (10.0.2.15)  0.057 ms  0.008 ms  0.006 ms
student@fedora:~$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.67.46
Name:   google.com
Address: 2404:6800:4007:805::200e
```

1.

```
PS C:\Users\DELL> ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::b0f:7aab:72d:9bb3%16
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

2.

```
PS C:\Users\DELL> ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : HDC0124094
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Ethernet Connection (17) I219-LM
   Physical Address. . . . . . . . . : 20-88-10-86-7A-BE
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

3.

```
PS C:\Users\DELL> ping HDC0124094

Pinging HDC0124094 [172.16.75.34] with 32 bytes of data:
Reply from 172.16.75.34: bytes=32 time<1ms TTL=128
Reply from 172.16.75.34: bytes=32 time<1ms TTL=128
Reply from 172.16.75.34: bytes=32 time<1ms TTL=128
Reply from 172.16.75.34: bytes=32 time<1ms TTL=128

Ping statistics for 172.16.75.34:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

4.

```
PS C:\Users\DELL> tracert HDC0124094
>>

Tracing route to HDC0124094 [172.16.75.34]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  HDC0124094 [172.16.75.34]
```

5.

```
Trace complete.
PS C:\Users\DELL> arp -a

Interface: 192.168.56.1 --- 0x10
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  224.77.77.77          01-00-5e-4d-4d-4d     static
  225.0.48.21           01-00-5e-00-30-15     static
```

6.

```
PS C:\Users\DELL> route print
===========================================================================
Interface List
  9...20 88 10 86 7a be ......Intel(R) Ethernet Connection (17) I219-LM
 16...0a 00 27 00 00 10 ......VirtualBox Host-Only Ethernet Adapter
 14...4e 82 a9 78 96 e7 ......Microsoft Wi-Fi Direct Virtual Adapter
  3...42 82 a9 78 96 e7 ......Microsoft Wi-Fi Direct Virtual Adapter #2
 18...4c 82 a9 78 96 e7 ......Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
  1...........................Software Loopback Interface 1
===========================================================================
```

7.

```
PS C:\Users\DELL> netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49672        HDC0124094:49673       ESTABLISHED
  TCP    127.0.0.1:49673        HDC0124094:49672       ESTABLISHED
  TCP    127.0.0.1:49674        HDC0124094:49675       ESTABLISHED
  TCP    127.0.0.1:49675        HDC0124094:49674       ESTABLISHED
  TCP    172.16.75.34:7680      172.16.75.11:53361     ESTABLISHED
  TCP    172.16.75.34:7680      172.16.75.19:52288     ESTABLISHED
```

8.

```
PS C:\Users\DELL> hostname
HDC0124094
```

9.

```
PS C:\Users\DELL> nslookup google.com
Server:  UnKnown
Address:  172.16.72.1

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4007:805::200e
          142.250.67.46
```